

Perceptron

OVERVIEW

Recall: linear classifiers

Inputs are d dimensional vectors, denoted by $\mathbf{x}$

Output is a label $y \in \{-1, 1\}$

Linear Threshold Units classify an example $\mathbf{x}$ using parameters $\mathbf{w}$ (a d dimensional vector) and b (a real number) according to the following classification rule

$$\text{Output} = \text{sign}(\mathbf{w}^T \mathbf{x} + b) = \text{sign}\left(\sum_i w_i x_i + b\right)$$

$$\text{if } \mathbf{w}^T \mathbf{x} + b \geq 0 \Rightarrow y = +1$$

$$\text{if } \mathbf{w}^T \mathbf{x} + b < 0 \Rightarrow y = -1$$

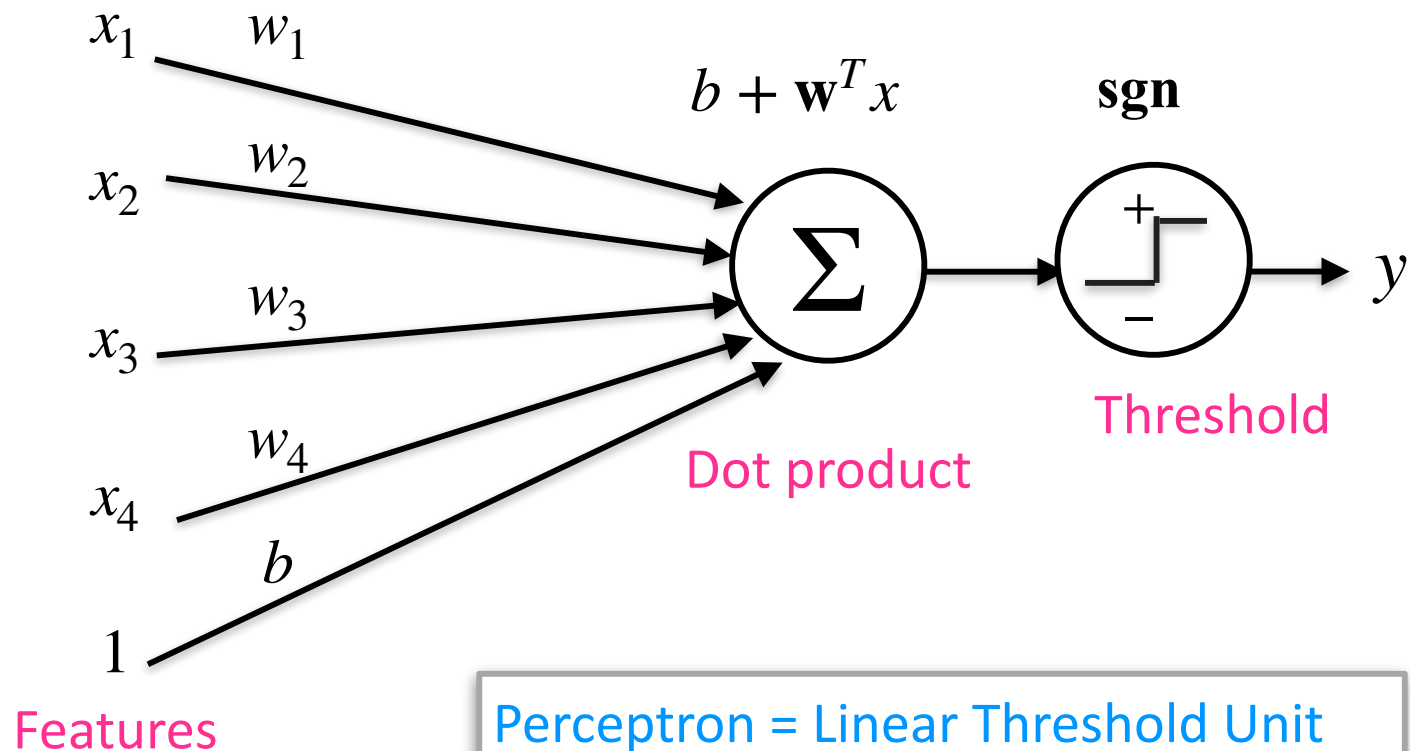
b is called the *bias* term

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Linear Threshold Functions

Many functions are Linear

- Conjunctions:

- $y = x_1 \wedge x_3 \wedge x_5$

- $y = \text{sgn}\{1 \cdot x_1 + 1 \cdot x_3 + 1 \cdot x_5 - 3\}; \quad \mathbf{w} = (1,0,1,0,1) \quad b = 3$

- At least m of n :

- $y = \text{at least } 2 \text{ of } \{x_1, x_3, x_5\}$

- $y = \text{sgn}\{1 \cdot x_1 + 1 \cdot x_3 + 1 \cdot x_5 - 2\}; \quad \mathbf{w} = (1,0,1,0,1) \quad b = 2$

Many functions are not

- Xor: $y = (x_1 \wedge x_2) \vee (\neg x_1 \wedge \neg x_2)$

- Non trivial DNF: $y = (x_1 \wedge x_2) \vee (x_3 \wedge x_4)$

But can be made linear

Note: all the variables above are Boolean variables

Dot product

The dot product of two vectors is written as $\mathbf{m}^T \mathbf{x}$ or $\mathbf{m} \cdot \mathbf{x}$, which

is defined as:
$$\mathbf{m}^T \mathbf{x} = \sum_{i=1}^k m_i x_i$$

Example

$$\mathbf{m} = \langle 5.13, 1.08, -0.03, 7.29 \rangle$$

$$\mathbf{x} = \langle x_1, x_2, x_3, x_4 \rangle$$

$$\mathbf{m}^T \mathbf{x} = 5.13x_1 + 1.08x_2 - 0.03x_3 + 7.29x_4$$

If dot product of two vectors is zero: means the two vectors are perpendicular (90° angle)

Length or norm of a vector

The length or norm of a vector $\mathbf{v}$ is the square root of length = $||\mathbf{v}|| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$

Dot product here is not zero ($\mathbf{v}$ is not perpendicular to itself), so now have 0° angle: dot product of $\mathbf{v} \cdot \mathbf{v}$ gives length of $\mathbf{v}$ squared

In 2 dimensions: length = $||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2}$

In 3 dimensions: length = $||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2 + v_e^2}$

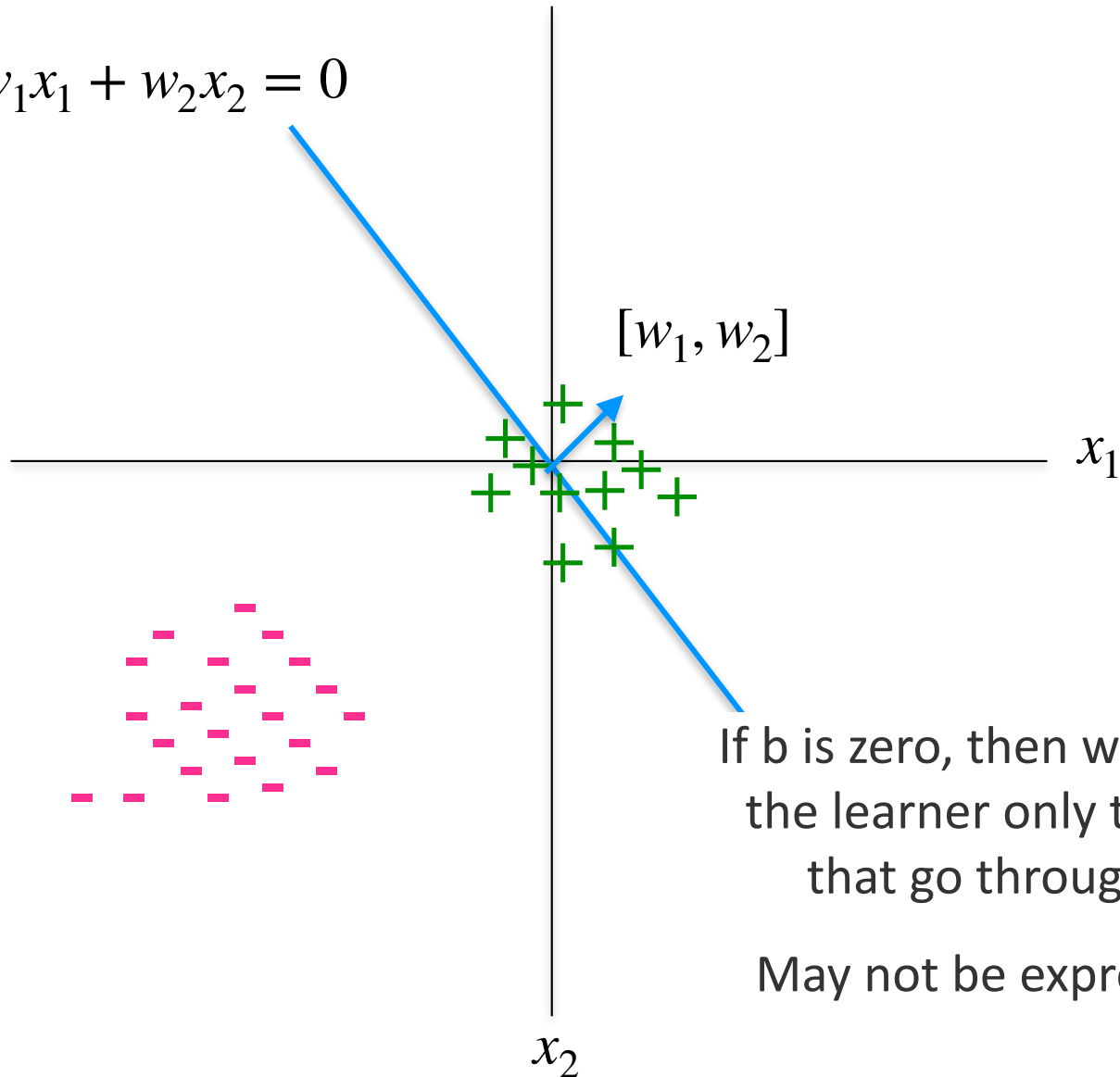
See Introduction to Linear Algebra by Gilbert Strang

Why is the bias term needed?

An illustration in two dimensions

$$\text{sgn}(b + w_1x_1 + w_2x_2)$$

$$w_1x_1 + w_2x_2 = 0$$



If b is zero, then we are restricting the learner only to hyperplanes that go through the origin

May not be expressive enough

Summary: perceptron

- Online learning algorithm, very widely used, easy to implement
- Additive updates to weights
- Geometric interpretation
- You should be able to implement the perceptron algorithm